

AOS-CX 10.08 Monitoring Guide

4100i, 6000, 6100 Switch Series



Copyright Information

© Copyright 2023 Hewlett Packard Enterprise Development LP.

Open Source Code

This product includes code licensed under the GNU General Public License, the GNU Lesser General Public License, and/or certain other open source licenses. A complete machine-readable copy of the source code corresponding to such code is available upon request. This offer is valid to anyone in receipt of this information and shall expire three years following the date of the final distribution of this product version by Hewlett Packard Enterprise Company. To obtain such source code, send a check or money order in the amount of US \$10.00 to:

Hewlett Packard Enterprise Company
6280 America Center Drive
San Jose, CA 95002
USA

Notices

The information contained herein is subject to change without notice. The only warranties for Hewlett Packard Enterprise products and services are set forth in the express warranty statements accompanying such products and services. Nothing herein should be construed as constituting an additional warranty. Hewlett Packard Enterprise shall not be liable for technical or editorial errors or omissions contained herein.

Confidential computer software. Valid license from Hewlett Packard Enterprise required for possession, use, or copying. Consistent with FAR 12.211 and 12.212, Commercial Computer Software, Computer Software Documentation, and Technical Data for Commercial Items are licensed to the U.S. Government under vendor's standard commercial license.

Links to third-party websites take you outside the Hewlett Packard Enterprise website. Hewlett Packard Enterprise has no control over and is not responsible for information outside the Hewlett Packard Enterprise website.

Acknowledgment

Intel®, Itanium®, Optane™, Pentium®, Xeon®, Intel Inside®, and the Intel Inside logo are trademarks of Intel Corporation in the U.S. and other countries.

Microsoft® and Windows® are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries.

Adobe® and Acrobat® are trademarks of Adobe Systems Incorporated.

Java® and Oracle® are registered trademarks of Oracle and/or its affiliates.

UNIX® is a registered trademark of The Open Group.

All third-party marks are property of their respective owners.

Contents	3
About this document	5
Applicable products	5
Latest version available online	5
Command syntax notation conventions	5
About the examples	6
Identifying switch ports and interfaces	6
Monitoring hardware through visual observation	8
Confirming normal operation of the switch by reading LEDs	8
Detecting if the switch is not ready for a failover event	9
Finding faulted components using the switch LEDs	9
Switch and port LEDs for 6000 Switch Series and 6100 Switch Series	11
Alarms	13
Alarm commands	13
alarm	13
alarm input	14
alarm snooze	15
show alarm	16
show alarm input	17
show alarm timer	18
Boot commands	20
boot set-default	20
boot system	20
show boot-history	22
Switch system and hardware commands	25
L1-100Mbps downshift	26
Limitations with speed downshift	26
L1-100Mbps downshift commands	26
downshift enable	26
show interface	27
show interface downshift-enable	29
show running-config interface	30
Mirroring	33
Mirror statistics	33
Classifier policies and mirroring sessions	33
Mirroring commands	34
clear mirror	34
comment	34

destination cpu	35
destination interface	36
diagnostic	37
disable	38
enable	39
mirror session	40
show mirror	40
source interface	42
source vlan	44
Monitoring a device using SNMP	46
Power-over-Ethernet	47
PoE commands	48
lldp dot3 poe	48
lldp med poe	49
power-over-ethernet	49
power-over-ethernet allocate-by	50
power-over-ethernet always-on	51
power-over-ethernet assigned-class	52
power-over-ethernet pre-std-detect	53
power-over-ethernet priority	54
power-over-ethernet quick-poe	54
power-over-ethernet threshold	55
power-over-ethernet trap	56
show lldp local	57
show lldp neighbor	58
show power-over-ethernet	59
Aruba AirWave	61
SNMP support and AirWave	61
SNMP on the switch	61
Supported features with AirWave and the AOS-CX switch	62
Configuring the AOS-CX switch to be monitored by AirWave	62
logging	63
snmp-server community	65
snmp-server host	66
snmp-server vrf	68
snmpv3 context	68
snmpv3 user	69
Support and Other Resources	72
Accessing Aruba Support	72
Accessing Updates	72
Aruba Support Portal	72
My Networking	73
Warranty Information	73
Regulatory Information	73
Documentation Feedback	73

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 4100i Switch Series (JL817A, JL818A)
- Aruba 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A)
- Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ <code><example-text></code> ■ <i><example-text></i> ■ <code>example-text</code> ■ <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.

Convention	Usage
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the interface context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>)#
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 4100i Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.

On the 6000 and 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.

Confirming normal operation of the switch by reading LEDs

This task describes using the switch LEDs to confirm that the switch is operating normally.

Procedure

1. Quick check: Verify that the chassis has power and there are no fault conditions.
On the front of the switch, verify that the states of the following LEDs are On Green:
 - Power
 - Health
2. Verify that the Health LEDs of all installed line modules are On Green.
3. Verify that the Health LEDs of all installed management modules are On Green.
4. Verify that the network ports are operating normally.
 - a. On the active management module, check the Status Front section. Verify that each LED that indicates a line module is in one of the following states:
 - On Green (normal operation)
 - Off (no line module installed)
 - b. On each line module, verify that each port LED is in one of the following states:
 - On Green, Half-Bright Green, or Flickering Green (normal operation)
 - Off (no cable connected or port off by default in config)
5. Verify that the power supplies are operating normally.
 - a. On the active management module, check the Status Front section. Verify that each LED that indicates a power supply is in one of the following states:
 - On Green (normal operation)
 - Off (no power supply installed)
 - b. On each power supply, verify that LEDs are in the following states:
 - Power LED: On Green
 - Fault LED: Off
6. Verify that the rear components are operating normally by checking the Status Rear section of the active management module:
 - a. Verify that the LEDs for the fabric modules are in one of the following states:
 - On Green (normal operation)
 - Off (component not installed)
 - b. Verify that the LEDs for the fan trays and fans are On Green.
7. Verify that the standby management module is ready to take over as the active management module. On the standby management module, verify the states of the following LEDs:

- Health LED is On Green.
- Management state standby (Stby) LED is On Green.

Detecting if the switch is not ready for a failover event

This task describes using the switch LEDs to detect if the switch is not ready for the loss of a fabric module or for a failover from the active management module to the standby management module.



Although you can detect power supply failures by viewing the LEDs, you must use software commands to determine if the power supply redundancy is sufficient to power the chassis if a power supply fails.

Procedure

1. Detect if the standby management module is shut down.
If the standby management module is shut down, the LED states are as follows:
 - The standby management module health LED is Off.
 - The standby management state active (Actv) LED is Off.
 - The standby management state standby (Stby) LED is Off.
 - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Off. For example, if the active management module is Management Module LED 5, Management Modules LED 6 is Off.
2. Detect if the standby management module is in a transient state. If the standby management module is booting, updating, or in another transient state, the LED states are as follows:
 - The standby management module health LED is Slow Flash Green when the service operating system is running or during an operating system update.
 - The standby management module Booting LED is Slow Flash Green when the AOS-CX operating system is booting.
 - The standby management state active (Actv) LED is Off.
 - The standby management state standby (Stby) LED is Off.
 - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Slow Flash Green.
3. Detect if a fabric module is shut down or not present. If a fabric module is shut down or not present, the LED states are as follows:
 - On the active management module, in the Status Rear section, the LED for the fabric module is Off.
 - On the rear display module, the LED for the fabric module is Off.
 - On the fabric module, the health LED is Off. However, the fabric module is behind fan 1 and is not directly visible.

Finding faulted components using the switch LEDs

This task describes using the switch LEDs to find components that are in a fault condition.



All green LEDs—except for chassis power LEDs and the Usr1 LED—are off when the LED mode is set to Light Faults (The Usr1 LED of the LED Mode section of the active management module is On Green and the default behavior for the Usr1 LED is being used.).

Procedure

1. Find the switch that has the fault condition, which is indicated by a chassis health LED in the state of Slow Flash Orange.

The chassis health LED is located on the front of the switch and on the rear panel of the switch.

2. If you are at the back of the switch, on the rear panel, look for LEDs that are in the Slow Flash Orange state:

The Status Rear area has LEDs for power supplies, fabric modules, fan trays, and fans. The number on the LED represents the unit number of the component.

If the only LED in a state of Slow Flash Orange is the Chassis health LED, go to the front of the switch.

3. At the front of the switch, on the active management module, look for LEDs that are in the Slow Flash Orange state:
 - The Status Front area has LEDs for power supplies, line and fabric modules, and management modules. The number on the LED indicates the slot number of the component.
 - The Status Rear area has LEDs for fabric modules and fan trays, with a single LED for all the fans in the fan tray. The number on the LED represents the slot or bay number of the component.

4. Use the number indicated by the LED that is flashing to locate the slot that contains the faulted component.

The fabric modules are located behind the fan trays, and the fabric module number corresponds to the fan tray number.

5. At the front of the switch, on line modules, look for LEDs that are in the Slow Flash Orange state:
Module LEDs and Port LEDs indicate faults if their states are Slow Flash Orange.

Chapter 3

Switch and port LEDs for 6000 Switch Series and 6100 Switch Series

Table 1: Switch and port LEDs

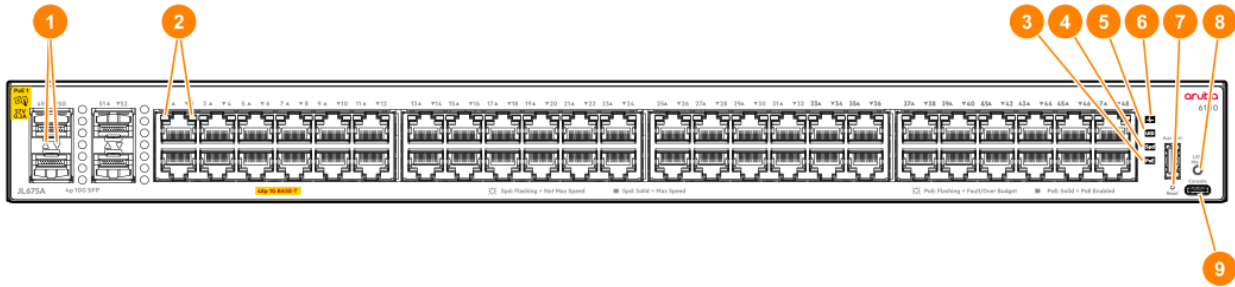


Table 2: Switch and port LEDs: Labels and description

Label	Description
1	Management Console LED
2	Switch port LEDs Back Module status LED
3	PoE mode selected
4	Speed mode selected LED
5	UID (Unit Identification) Reset button Usr mode selected LED
6	Global Status LED
7	Reset button
8	LED Mode status LED
9	USB-C Console port

Table 3: Front Panel LED behavior

Switch LEDs	Function	State	Meaning
Back LED	Status of modular components installed in the back of the chassis (not applicable for 6200F/6300F switches)	On - Green	Normal
		Slow Flash - Amber	Fault in one of the modules in the back of the chassis

Switch LEDs	Function	State	Meaning
PoE LED	Indicates Port LEDs are showing PoE information (not applicable for non PoE switches)	Off	PoE mode not selected
		On - Green	PoE mode selected
		Slow Flash - Amber	Hardware failure PoE enabled port, PoE mode not selected
		On - Amber	Hardware failure PoE enabled port, PoE mode selected
Spd LED	Indicates Port LEDs are showing speed information	Off	Speed mode not selected
		On - Green	Speed mode selected
UID LED	User-configurable LED	Off	User defined the located LED : OFF
		On/Flash Blue (for 30 min)	User defined the locator LED: On/Flash
Global Status Indicator LED	Overall status of the product	Flash - Green	Self-test in progress during UBOOT, SVOS and AOS-CX
		On - Green	Successfully initialized AOS-CX
		Flash - Amber	Recoverable faults (e.g. fans, PSU fault)
		On - Amber	Critical faults (e.g. exceed temperature limit)
* Press the Mode Select button to switch between User(default), PoE, or Spd.			

Table 4: *Rear Panel LED behavior*

Switch LEDs	Function	State/Mode	Meaning
Fan health LED	Status of fan	On - Green	Normal
		Slow flash - Amber	Fan fault
UID LED	User-configurable LED	Off	User define the locator LED : OFF
		On/Flash (30 min) - blue	User define the locator LED: On/Flash
PSU Status Indicator LED	Status of power supply	On Green	Normal
		Off	No power, PSU has invalid AC input of invalid DC outputs
		Slow Flash - Green	Power supply has faulted or warning

Alarms allow you to configure physical alarms for alarm input ports and specific systems events.



Alarms are only available on the 4100i switch.

Alarm commands

alarm

```
alarm {temperature | power supply} [action <LOG-AND-TRAP> | <RELAY>]  
no alarm {temperature | power supply} [action <LOG-AND-TRAP> | <RELAY>]
```

Description

Configures global events to be forwarded to the output alarm port. The `no` form of this command disables the specified configuration.

Parameter	Description
{temperature}	Selects the alarm for ambient temperatures reaching threshold limits. The threshold is 70°C.
{power supply}	Selects the alarm events from the power supply.
<LOG-AND-TRAP>	Generates an event log and SNMP trap.
<RELAY>	Relays an event to alarm output port.

Examples

Disabling a temperature event to remove the configuration for all actions associated with the event:

```
switch(config)# no alarm temperature
```

Configuring an alarm for a temperature event for the log-and-trap action:

```
switch(config)# alarm temperature action log-and-trap
```

Removing the configuration for the temperature event for a log-and-trap action:

```
switch(config)# no alarm temperature action log-and-trap
```

Configuring an alarm for a power-supply event for the relay action:

```
switch(config)# alarm power-supply action relay
```

Removing the configuration for the power-supply event for the relay action:

```
switch(config)# no alarm power-supply action relay
```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

alarm input

```
alarm input {IN1 | IN2} [name <STRING>] [action <LOG-AND-TRAP> | <RELAY>] [trigger  
<CLOSED | OPEN>]  
no alarm input {IN1 | IN2} [name <STRING>] [action <LOG-AND-TRAP> | <RELAY>] [trigger  
<CLOSED | OPEN>]
```

Description

Configures input alarm and actions. The `no` form of this command removes the specified configuration.

Parameter	Description
{IN1 IN2}	Specifies the input alarm port..
name	Specifies the external device connected.
<STRING>	Descriptive string.
action	Specifies the action to be taken when the monitored event occurs.
<LOG-AND-TRAP>	Generates an event log and SNMP trap.
<RELAY>	Relays an event to alarm output port.
trigger	Triggers an alarm based on a normally open or closed circuit.
<CLOSED>	Generates an alarm event when the circuit is closed.

Parameter	Description
<OPEN>	Generates an alarm event when the circuit is open.

Examples

Configuring an alarm on input port 1 named Door-Sensor:

```
switch(config)# alarm input IN1 name Door-Sensor
```

Removing the configuring for an alarm on input port 1:

```
switch(config)# no alarm input IN1
```

Configuring an alarm on input port 1 with log-and-trap action:

```
switch(config)# alarm input IN1 action log-and-trap
```

Removing the configuration for an alarm on input port 1 with log-and-trap action:

```
switch(config)# no alarm input IN1 action log-and-trap
```

Configuring an alarm on input port 1 to trigger an alarm when the door sensor circuit is closed:

```
switch(config)# alarm input IN1 trigger closed
```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

alarm snooze

```
alarm snooze [time in minutes] [repeat]
```

Description

Configures any active event forwarded to alarm relay to be snoozed.

Parameter	Description
time in minutes	Specifies the value for time in minutes. The range is 0-1440 minutes..
repeat	Repeats the previous snooze time.

Examples

Configuring an alarm relay action for 10 minutes:

```
switch(config)# alarm snooze 10
```

Configuring an alarm relay action to be repeated with previously configured snooze time:

```
switch(config)# alarm snooze repeat
```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

show alarm

```
show alarm [temperature | power supply]
```

Description

Shows all of the details of global status monitoring alarm events.

Parameter	Description
temperature	Selects the alarm for ambient temperatures reaching threshold limits.
power supply	Selects the alarm events from the power supply.

Examples

Showing details for all global alarm events on the switch:

```
switch# show alarm
Alarm Snooze Timer Status: active
Duration remaining: 1 min 32 sec
```



```

-----
Global Alarm: Temperature
-----
Alarm Event      Status      log-and-trap  Relay
-----
Temperature      inactive    true           false

-----
Global Alarm: power-supply
-----
Alarm Event      Status      log-and-trap  Relay
-----
power-supply     active      false          true

```

Showing details for the temperature alarm:

```

switch# show alarm temperature
Alarm Snooze Timer Status: Active
Duration remaining: 5 min 36 sec
-----
Global Alarm: Temperature
-----
Alarm Event      Status      log-and-trap  Relay
-----
Temperature      inactive    true           false

```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	Manager (#)	Administrators or local user group members with execution rights for this command.

show alarm input

```
show alarm input [IN1 | IN2]
```

Description

Shows the alarm details for all ports or for the specified port.

Parameter	Description
{IN1 IN2}	Specifies the input alarm port..

Examples

Showing details for all alarm input ports on the switch:

```

switch# show alarm input
Alarm Snooze Timer Status: inactive
-----
Input Alarm IN1, Name: Door-Sensor
-----
Alarm Port      Status      log-and-trap  Relay      Trigger
-----
IN1             inactive   true          false      closed

-----
Input Alarm IN2, Name: N/A
-----
Alarm Port      Status      log-and-trap  Relay      Trigger
-----
IN2             active     false         true       open

```

Showing details for alarm input ports IN1:

```

switch# show alarm input IN1
Alarm Snooze Timer Status: inactive
-----
Input Alarm IN1, Name: Door-Sensor
-----
Alarm Port      Status      log-and-trap  Relay      Trigger
-----
IN1             inactive   true          false      closed

```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	Manager (#)	Administrators or local user group members with execution rights for this command.

show alarm timer

```
show alarm timer
```

Description

Shows the status of an alarm snooze timer's status and duration.

Examples

Showing the status of an alarm snooze timer when it is inactive :

```

switch# show alarm timer
Alarm Snooze Timer Status: inactive

```

Showing the status of an active 3-minute alarm snooze timer:

```
switch# show alarm timer
Alarm Snooze Timer Status: active
Duration remaining: 2 min 55 sec
```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	Manager (#)	Administrators or local user group members with execution rights for this command.

boot set-default

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
primary	Selects the primary network operating system image.
secondary	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
<code>primary</code>	Selects the primary operating system image for this reboot and sets the configured default operating system image to <code>primary</code> for future reboots.
<code>secondary</code>	Selects the secondary operating system image for this reboot and sets the configured default operating system image to <code>secondary</code> for future reboots.
<code>serviceos</code>	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the `show images` command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the `primary` or `secondary` optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

You can use the `boot set-default` command to change the configured default operating system image.

- If you select `serviceos` as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the `boot system` command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the `boot system` command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

```
show boot-history [all]
```

Description

Shows boot information. When no parameters are specified, shows the most recent information about the boot operation, and the three previous boot operations for the active management module. When the `all` parameter is specified, shows the boot information for the active management module.

Parameter	Description
<code>all</code>	Shows boot information for the active management module and all available line modules.

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

Index

The position of the boot in the history file. Range: 0 to3.

Boot ID

A unique ID for the boot . A system-generated 128-bit string.

Current Boot, up for *<SECONDS>* seconds

For the current boot, the `show boot-history` command shows the number of seconds the module has been running on the current software.

Timestamp boot reason

For previous boot operations, the `show boot-history` command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values:

<DAEMON-NAME> crash

The daemon identified by *<DAEMON-NAME>* caused the module to boot.

Kernel crash

The operating system software associated with the module caused the module to boot.

Reboot requested through database

The reboot occurred because of a request made through the CLI or other API.

Uncontrolled reboot

The reason for the reboot is not known.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Chapter 6

Switch system and hardware commands

Switch system and hardware commands are general commands used to configure fundamental settings on the switch.



Refer to the Fundamentals Guide to view the switch system and hardware commands.

The speed downshift feature allows the user to link-up at sub-optimal speeds when failing to link-up at the highest advertised speed. There are fixed number of link attempts made to establish link at highest advertised speed and when all of them fail and attempt is made to link-up at a lower possible speed.

This feature requires underlying PHY to have support for the same and hence capability is only added to select set of ports. If a link cannot be established at the highest common denominator within a set number of link attempts, the PHY advertises the next highest speed using auto-negotiation.

Limitations with speed downshift

- Link up may be delayed as certain number of retries are done to establish the link at highest advertise speeds by both link partners before downshifting.
- Link may be established at sub-optimal speed.

L1-100Mbps downshift commands

downshift enable

```
downshift-enable  
no downshift-enable
```

Description

Enables/disables automatic speed downshift on an interface that supports downshift, generally 1GBASE-T ports. When enabled, downshift allows an interface to link at a lower advertised speed when unable to establish a stable link at the maximum speed. Downshifting only applies to physical interfaces that are not members of a LAG and is only available when auto-negotiation is enabled. When only one speed is advertised, downshift will not be triggered.

Examples

```
switch(config-if)# interface 1/1/1  
switch(config-if)# downshift-enable
```

```
Warning: this is a non-standard mode for use only when standards-based  
auto-negotiation is not able to establish a stable link. Enabling this  
may cause the port to link at a lower than expected speed and should  
not be used on ports that are members of a LAG. Support calls may require  
this feature to be disabled
```

```
Continue (y/n)?
```

```
switch(config-if)#
```

When automatic downshift is enabled:

```
switch(config-if)# show running-config interface  
interface 1/1/1  
    downshift-enable
```

Disabling automatic speed downshift:

```
switch(config-if)# interface 1/1/1  
switch(config-if)# no downshift-enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

show interface

```
show interface [<IFNAME>|<IFRANGE>] [brief | physical | extended [non-zero]]  
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief | physical]  
show interface [lag | loopback | tunnel | vlan ] [<ID>] [extended | non-zero]
```

Description

Displays active configurations and operational status information for interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
brief	Displays brief info in tabular format.
extended	Displays the physical connection info in tabular format.
non-zero	Displays only non zero statistics.
LAG	Displays LAG interface information.
LOOPBACK	Displays loopback interface information.
TUNNEL	Displays tunnel interface information.
VLAN	Displays VLAN interface information.

Parameter	Description
<LAG-ID>	Specifies the LAG number. Range: 1-256
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0-255
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1-255
<VLAN-ID>	Specifies the VLAN ID. Range: 1-4094
VXLAN	Displays the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1

Examples

Showing interfaces configured for L2 forwarding:

```
switch(config-if)# show interface
Interface 1/1/1 is down (Administratively down)
Admin state is down
State information: Administratively down
Link transitions: 2
Description: Backup data center link
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 9198
Type SFP-SX
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is on
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rx
    1386055 total packets          586397526 total bytes
    1374287 unicast packets
    11764  multicast packets
    4      broadcast packets
    0 errors                      0 dropped
    0 CRC/FCS
Tx
    2475612 total packets          1003506711 total bytes
    2311806 unicast packets
    100369  multicast packets
    63437  broadcast packets
    0 errors                      2462773 dropped
    0 collision

Interface 1/2/1 is down (Administratively down)
Admin state is down
State information: Administratively down
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 9198
Type QSFP+SR4
Full-duplex
qos trust none
Speed 0 Mb/s
```

```

Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rx
    0 total packets                0 total bytes
    0 unicast packets
    0 multicast packets
    0 broadcast packets
    0 errors                        0 dropped
    0 CRC/FCS
Tx
    0 total packets                0 total bytes
    0 unicast packets
    0 multicast packets
    0 broadcast packets
    0 errors                        0 dropped
    0 collision

```

When the interface is currently linked at a downshifted speed:

```

switch(config-if)# show interface 1/1/1

Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface downshift-enable

```
show interface [<IFNAME>|<IFRANGE>] downshift-enable
```

Description

Displays speed downshift information, including the interface speed status and configuration.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.

Examples

Showing automatic downshift information:

```
switch(config-if)# show interface downshift-enable
```

Port	Downshift		Speed	
	Enabled	Active	Status	Config
1/1/1	yes	yes	100M-FDx	auto
1/1/2	yes	no	1G	auto
1/1/3	yes	no	100M-FDx	100M-FDx
1/1/4	no	no	--	auto

Showing automatic downshift information on per interface:

```
switch(config-if)# show interface 1/1/2 downshift-enable
```

Port	Downshift		Speed	
	Enabled	Active	Status	Config
1/1/2	yes	no	1G	auto

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show running-config interface

```
show running-config interface [<IFNAME>|<IFRANGE>]  
show running-config interface [lag | loopback | tunnel | vlan ] [<ID>]
```

Description

Displays active configurations of various switch interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
LAG	Specifies LAG interface information

Parameter	Description
LOOPBACK	Specifies loopback interface information.
TUNNEL	Specifies tunnel interface information.
VLAN	Specifies VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256.
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0-255.
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1-255.
<VLAN-ID>	Specifies the VLAN ID. Range: 1-4094.
VXLAN	Specifies the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1.

Examples

Showing 1/1/2 interface configuration:

```
switch(config-if)# show running-config interface 1/1/2

interface 1/1/2
  no shutdown
  description DC-23
  exit
```

Showing loopback interfaces configured:

```
switch(config-if)# show running-config interface loopback

interface loopback 1
  description lb interface 1
  exit
interface loopback 2
  description lb interface 2
  exit
```

Showing loopback interfaces not configured:

```
switch(config-if)# show running-config interface loopback

No loopback interfaces configured.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Mirroring allows you to replicate all traffic arriving and/or leaving the selected system interfaces. This data can be used for collection or analysis.

The traffic replicated using mirroring can be sent to a separate interface on the same switch as the traffic source for analysis or inspection. Such a collection of interfaces and settings is called a mirror session.

A mirror session can be configured with many traffic sources but only a single output, or destination. In the initial configuration, the mirror session is disabled. You have enable the feature to start the replication.



Care must be taken in choosing the number and rates of sources to avoid over-saturating a session destination. A mirror session with multiple 10G sources can overwhelm a single 10G destination and important data may be lost.

Mirror statistics

Mirror statistics are reset for a Mirror-to-CPU session when an interface is added or removed from a LAG that is a source interface in the Mirror session and during a failover.

Classifier policies and mirroring sessions

Network traffic can be mirrored to a destination interface in two ways:

- Using a mirroring session alone.
- Using Classifier Policies with mirror actions in conjunction with a mirroring session.

Basic mirroring sessions provide coarse control over the type of traffic mirrored from a source: all received, all transmitted, or both. However, a traffic class within a Classifier Policy applied to a source can provide much finer grained control of mirrored traffic. For example, a policy can match on many different aspects of the Ethernet or IPv4 or IPv6 header information in each frame or packet received or transmitted on an interface.

The steps to configure a policy and class with a mirror action are the following:

1. Configuring a mirroring session with a destination interface.
2. Enabling the mirroring session.
3. Configuring the Classifier Policy, specifying the mirroring session ID in the mirror action.

If the packets being mirrored are received from a VLAN that is not allowed on the mirror destination, the mirrored packets would be dropped at the mirror destination interface. When the mirrored packets are dropped at the destination, the mirror output packet and byte count will increment, however the packets will not be received at the mirror destination.

The mirror destination port among the active mirror sessions must be unique. That is, if an interface is configured as a source or destination in an active mirror session, the same port cannot be used as a destination in another active mirror session.

Mirroring commands

clear mirror

```
clear mirror [all | <SESSION-ID>]
```

Description

Clears the mirror statistics for all configured mirror sessions or a specified session

Parameter	Description
all	Specifies all configured sessions.
<SESSION-ID>	Specifies a numeric identifier for the session. Range: 1 to 4

Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

comment

```
comment <COMMENT>  
no comment
```

Description

Specifies a comment for the mirroring session.

When used in mirror endpoint command context, specifies a comment for the mirror endpoint.

The `no` form of this command removes the comment.

Parameter	Description
<code><COMMENT></code>	A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3) # comment This Mirror will be removed during next
maintenance window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3) # no comment
```

Adding a comment to a mirror endpoint:

```
switch(config-mirror-endpoint-test) # comment Monitor endpoint traffic
```

Replacing the existing comment for mirror endpoint:

```
switch(config-mirror-endpoint-test) # comment Monitor statistics on each endpoint
interfaces
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror-<SESSION-ID></code> <code>config-mirror-endpoint</code>	Administrators or local user group members with execution rights for this command.

destination cpu

```
destination cpu
no destination cpu
```

Description

The command causes the mirror session to transmit mirrored packets to the switch CPU. This destination may be configured for multiple sessions, however only one such configured session may be active at a given time.

The diagnostic utility Tshark may be used to view and capture packets transmitted to the CPU through this route. Ctrl+C must be entered to terminate a Tshark capture session. More details can be found in the *Supportability Guide*.

The `no` form of this command will immediately stops mirroring traffic to the CPU, but will not remove any sources from the mirror configuration.

Examples

Configuring a mirror session with CPU as the destination.

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination cpu
```

Removing the destination entirely.

```
switch(config-mirror-1)# no destination cpu
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

destination interface

```
destination interface {<INTERFACE-ID>|<LAG-NAME>}
no destination interface {<INTERFACE-ID>|<LAG-NAME>}
```

Description

Configures the specified interface as the destination of the mirrored traffic.

The `no` form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

Parameter	Description
<INTERFACE-ID>	Specifies a interface. Format: member/slot/port.
<LAG-NAME>	Specifies a LAG (link aggregation group) identifier.

Usage

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration.

Examples

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1  
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface 1/1/12
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

diagnostic

diagnostic

```
diag utilities tshark [file]  
diag utilities tshark [delete-file]
```

Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.



The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the `delete-file` form of this command to delete the most recent capture file.

Since `file` and `delete-file` are optional, the behavior of the base command `diag utilities tshark` does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

Parameter	Description
file	Saves captured packets to a temporary file.
delete-file	Deletes the most recent captured file.

Example

Performing diagnostic:

```
switch# diagnostic

switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

disable

disable

Description

Disables the mirroring session specified by the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the `show mirror` command for that session ID shows an Admin Status of `disable` and an Operation Status of `disabled`.

Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror-<SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

enable

enable

Description

Enables the mirroring session for the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the `show mirror` command for that session ID shows an `Admin Status` of `enable` and an `Operation Status` of `enabled`.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of `rx` or `both`):

- The attempt to enable the mirroring session fails and an error is returned.



When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

Example

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# source interface 1/1/2 rx
switch(config-mirror-3)# destination interface 1/1/3
switch(config-mirror-3)# comment Monitor router port ingress-only traffic
switch(config-mirror-3)# enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

mirror session

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The `no` form of this command removes an existing mirroring session from the configuration.

Parameter	Description
<SESSION-ID>	Specifies the session identifier. Range: 1 to 4

Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#

switch(config)# mirror session 3
switch(config-mirror-3)#

switch(config)# no mirror session 1
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show mirror

```
show mirror [<SESSION-ID>]
```

Description

Shows information about mirroring sessions. If <SESSION-ID> is not specified, then the command shows a summary of all configured mirroring sessions. If <SESSION-ID> is specified, then the command shows detailed information about the specified mirroring session.

Parameter	Description
<SESSION-ID>	Specifies the session identifier. Range: 1 to 4

Usage

Admin Status indicates the configured status. Admin Status is one of the following values:

enable

The mirroring session is enabled.

disable

The mirroring session has been configured but not yet enabled, or has been disabled.

Operation Status indicates the status of the mirroring session. Operation Status is one of the following values:

dest_doesnt_exist

The configured destination interface is not found in the system. The mirroring session cannot be enabled.

destination_shutdown

The mirroring session is enabled, but the destination interface is shut down. No traffic can be monitored.

disabled

The mirroring session is disabled and is not in an error condition.

enabled

The mirroring session is enabled.

external/driver_error

An internal ASIC hardware error occurred.

hit_active_sessions_capacity

The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.

internal_error

An invalid parameter was passed to the ASIC software layer.

no_dest_configured

The mirroring session does not have a destination interface configured.

no_name_configured

A software error occurred. The mirroring session does not have a session ID in its configuration.

null_mirror

A software error occurred. The session object reference is invalid.

out_of_memory

The system is out of memory, reboot recommended.

tunnel_route_resolution_not_populated

If the destination tunnel IP address is not reachable.

unknown_error

An unexpected error occurred.

Examples

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID  Admin Status  Operation Status
---  -
1   enable       enabled
2   disable      disabled
3   disable      disabled
4   enable       internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
Mirror Session: 3
Admin Status: disable
Operation Status: disabled
Comment: Monitor router port ingress-only traffic
Source: interface 1/1/2 rx
Destination: interface 1/1/3
Output Packets: 0
Output Bytes: 0
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

source interface

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored. The **no** form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

Parameter	Description
<PORT-NUM>	Specifies a physical port on the switch. Use the format member/slot/port (for example, 1/3/1).
<LAG-NAME>	Specifies the identifier for the LAG (link aggregation group).
<DIRECTION>	Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following:
both	Mirror both transmitted and received packets.
rx	Mirror only received packets.
tx	Mirror only transmitted packets.

Usage

There is a limit of source interfaces in each direction of a given mirror session:

Switch	Source interface limit
4100i	4
6000	4
6100	4

However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.



When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1) # source interface
LAG-NAME      Enter a LAG name. For example, lag10
PORT-NUM      Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config) # mirror session 1
switch(config-mirror-1) # source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config) # mirror session 2
switch(config-mirror-2) # source interface 1/1/3 tx
switch(config-mirror-2) # source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2) # no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3) # source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1) # source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4) # source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4) # no source interface lag1 rx
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror- <i><SESSION-ID></i>	Administrators or local user group members with execution rights for this command.

source vlan

Syntax

```
source vlan <VLAN-NUM> {rx | tx | both}
no source vlan <VLAN-NUM> [rx | tx | both]
```

Description

Adds or removes VLAN as a source of traffic to be mirrored. More than one source VLAN can be configured in a mirror session. Each VLAN may specify its own direction.

The `no` version of the command ceases mirroring traffic from the specified source VLAN and removes the source from the mirror configuration.

There is a limit of 1024 source VLANs in each direction of a given mirror session. The same VLAN can be configured as a mirror source for multiple sessions.

Command context

config

Parameters

<VLAN-NUM>

Configured VLAN number.

rx

Mirror only received traffic.

tx

Mirror only transmitted traffic.

both

Mirror both received and transmitted traffic.

Authority

Administrators or local user group members with execution rights for this command.

Example

Create a mirror session and add VLAN 10 as a source of traffic in both directions on that port.

```
switch(config)# mirror session 1  
switch(config-mirror-1)# source vlan 10 both
```

Create a second mirror session and add VLAN 10 as a transmit sources of traffic and VLAN 20 in both receive and transmit directions.

```
switch(config)# mirror session 2  
switch(config-mirror-2)# source vlan 10 tx  
switch(config-mirror-2)# source vlan 20 both
```

Reconfigure the source in session 2 to be receive only by respecifying the source interface configuration.

```
switch(config-mirror-2)# source vlan 10 rx
```

From the second session, remove the first source interface entirely and remove the transmit direction from the other so that mirroring only occurs in the receive direction.

```
switch(config-mirror-2)# no source vlan 10  
switch(config-mirror-2)# no source vlan 20 tx
```

Message received when trying to add more than 1024 mirror source VLANs

```
switch(config-mirror-2)# source vlan 2000 rx  
The maximum number of source VLANs per mirror session is 1024 in each direction
```

Chapter 9

Monitoring a device using SNMP

Configuring SNMP: Refer to the *SNMP/MIB Guide* for information on how to add SNMP so a device can be monitored from a network management system (NMS).

Configuring an SNMP trap receiver: Refer to the *SNMP/MIB Guide* and specific information about the `show snmp trap` command to enable SNMP traps.

- The Power-over-Ethernet (PoE) subsystem manages power supplied to devices using standard Ethernet data cables. A Power Sourcing Equipment (PSE) supplies DC power as well as Ethernet connectivity to a Powered Device (PD) using a standard Ethernet cable. The maximum current depends on the PD Requested Class.
- A PoE subsystem contains two parts : a PSE and PD. A Power Sourcing Equipment (PSE) is a device that provides power through a standard Ethernet cable. A PoE capable switch functions as PSE. All Aruba PoE switches are considered as PSEs. A PD is a device powered by a PSE. Examples of PD are VoIP phones, Wireless APs, and IP cameras.
- When a PD or any network cable is connected to a PSE port, the PSE applies a detection voltage and measures the resistance value of the PD. If resistance is within IEEE 802.3 standard values (23 - 26k ohm), the connected device is treated as PD and classification begins. For legacy devices to be detected, you must enable prestandard detection on the switch.
- PDs are divided into different types and classes based on PD power requirements. The power supplied by the PSE is higher than the power PD draws to accommodate for the line losses that can result with the use of the standard maximum length cable(100m).
 - Type 1: PSE can supply maximum of 15.4W, and PD can draw a maximum of 13W.
 - Type 2: PSE can supply maximum of 30W, and PD can draw a maximum of 25.5W.
 - Type 3: PSE can supply maximum of 60W, and PD can draw a maximum of 51W.
 - Type 4: PSE can supply maximum of 90W, and PD can draw a maximum of 71W.
- Classes of PD:
 - Class 0: Type1 PD, it can draw a maximum of 13W.
 - Class 1: Type1 PD, it can draw a maximum of 3.84W.
 - Class 2: Type1 PD, it can draw a maximum of 6.49W.
 - Class 3: Type1 PD, it can draw a maximum of 13W.
 - Class 4: Type2 PD, it can draw a maximum of 25.5W.
 - Class 5: Type3 PD, it can draw a maximum of 40W.
 - Class 6: Type3 PD, it can draw a maximum of 51W.
 - Class 7: Type4 PD, it can draw a maximum of 62W.
 - Class 8: Type4 PD, it can draw a maximum of 71.3W.
- IEEE 802.3bt introduced 4-Pair PoE as a means of supplying higher power to PDs that need more than the current 25.5W supplied by IEEE 802.3at. To increase the available power without damaging the Ethernet cable, the standard introduced the ability to use all four pairs within the Ethernet cable instead of the two pairs used by previous standards (802.3at, 802.3af).
- Supported protocols:
 - Compatibility with IEEE 802.3af, 802.3at, 802.3bt and prestandard.
 - Long first class event supported on Type 3-4 PSE.
 - Support for Single Signature (SS) Type 0-6 and Dual Signature (DS) Type 0-4 PDs.
 - Multi-Event classification permits mutual ID of SS Class 0-6 and DS Class 0-4.

- Support LLDP Data Link Layer (DLL) Type 1-2 extension 12-octet TLV and Type 3-4 extension 29-octet TLV.
- Default PSE assigned class delivers the maximum PSE capable power at initial power up based on PD requested class.
- Always-on PoE is a feature that provides the ability for a switch to continue to provide power across user initiated reboots through software. Always-on PoE is enabled by default and no additional configuration is needed.



PDs only remain powered, no data transfer or PoE power negotiation can occur until the switch has completely booted up and in normal operation. PD faults occurring prior to full switch boot up will result in PoE power removal and restart the detection process only after switch returns to normal operation.

PoE commands

All PoE configuration commands except `threshold configuration` and `always-on poe` configuration are entered at the `config-if` context. The PoE threshold command is used at the system level whereas the `always-on poe` and `power-over-ethernet quick-poe` commands are set at the slot level. These commands can only be configured in the global configuration context.

lldp dot3 poe

```
lldp dot3 poe
no lldp dot3 poe
```

Description

Enables 802.3 TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. LLDP dot3 TLV is by default enabled for PoE.

The `no` form of this command disables 802.3 TLV list in LLDP.

Examples

Enabling 802.3 TLV list in LLDP:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 poe
```

Disabling 802.3 TLV list in LLDP:

```
switch(config-if)# no lldp dot3 poe
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

lldp med poe

```
lldp med poe [priority-override]
no lldp med poe [priority-override]
```

Description

Enables MED TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. Also enables the lldp-MED TLV priority to override user configured port priority for Power over Ethernet. When both dot3 and MED are enabled, dot 3 will take precedence. MED TLV is by default enabled for PoE. Priority over-ride is by default disabled.

The `no` form of this command disables MED TLV list in LLDP.

Parameter	Description
[priority-override]	System defined name of the interface.

Examples

Enabling and disabling LLDP MED PoE:

```
switch(config) # interface 1/1/1
switch(config-if) # lldp med poe
switch(config-if) # no lldp med poe
```

Enabling and disabling LLDP MED PoE priority override:

```
switch(config-if) # lldp med poe priority-override
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet

```
power-over-ethernet
no power-over-ethernet
```

Description

Enables per-interface power distribution. Per-port power is enabled by default with priority low. PoE cannot be disabled for individual ports when Quick PoE is enabled for the entire switch or line module. The `no` form of this command disables per-interface power distribution.

Examples

Enabling per-interface power distribution:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet
```

Disabling per-interface power distribution:

```
switch(config-if)# no power-over-ethernet
```

Showing Quick PoE enabled:

```
switch(config-if)# power-over-ethernet quick-poe 1/1
switch(config-if)# interface 1/1/1
switch(config-if)# no power-over-ethernet
Interface PoE cannot be disabled when Quick PoE is enabled.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet allocate-by

```
power-over-ethernet allocate-by {usage | class}
no power-over-ethernet allocate-by {usage | class}
```

Description

Configures the power allocation method. Power allocation method is initially based on usage. PSE Allocated power value will change to LLDP negotiated power if and when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocated Power Value will be the actual instantaneous power draw and reserve power based on actual consumption. In allocate-by class, power allocation is based on PD requested class and PSE allocated power value will be the LLDP negotiated power when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocate Power will be based on PD class. Reserve power is based on PD Class. By default, power allocation is by usage.

The `no` form of this command resets the action to default.

Examples

Configuring the power allocation method:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet allocate-by usage
switch(config-if)# power-over-ethernet allocate-by class
```

Resetting power allocation method:

```
switch(config-if)# no power-over-ethernet allocate-by class
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet always-on

```
power-over-ethernet always-on <MODULE-ID>
no power-over-ethernet always-on <MODULE-ID>
```



This command is not available on the 6000 or 6100 Switch Series.



This command is available only on the 4100i Switch Series.

Description

Always-on PoE is a feature that provides the ability to the switch to continue to provide power across a soft reboot. It is applicable only to the interfaces which were connected and delivering before the soft reboot. Also, power will not be delivered if power to the switch is interrupted. This command enables or disables the always-on PoE feature at the switch or the slot level. By default, always-on PoE is enabled at the switch or the slot level.

The `no` form of this command disables power distribution on soft reboot.

Parameter	Description
<MODULE-ID>	Module number to apply always-on PoE configuration.

Examples

Enabling per-interface power distribution:

```
switch(config)# power-over-ethernet always-on 1/1
```

Disabling per-interface power distribution:

```
switch(config)# no power-over-ethernet always-on 1/1
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

power-over-ethernet assigned-class

```
power-over-ethernet assigned-class {3 | 4 | 6}  
no power-over-ethernet assigned-class
```

Description

Limit PoE power based on the assigned class. When an user assigns a maximum class to an interface, the PSE will limit the maximum power delivered to the PD up to a total power draw not exceeding the PSE assigned-class power. Power demotion occurs when a PD requested class is higher than the PSE assigned class, permitting the PD to receive power and operate in a reduced power mode. PoE ports cannot set an assigned class when Quick PoE is enabled on the subsystem. The default assigned class is 4 for 2-pair capable PSE and 6 for 4-pair capable PSE.

The `no` form of this command resets the action to default.

Examples

Setting PoE assigned class:

```
switch(config)# interface 1/1/1  
switch(config-if)# power-over-ethernet assigned-class 4
```

Resetting PoE assigned class to default:

```
switch(config-if)# no power-over-ethernet assigned-class 4
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# interface 1/1/1
switch(config)# power-over-ethernet assigned-class 4
Interface assigned class cannot be configured when Quick PoE is enabled.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet pre-std-detect

```
power-over-ethernet pre-std-detect
no power-over-ethernet pre-std-detect
```

Description

Before IEEE 802.3 released the first Power over Ethernet standard (802.3af), vendors had shipped PoE capable switches and PD's. As we are backward compatible Aruba will support both IEEE standard and pre-standard 802.3af Power over Ethernet PD's concurrently. This CLI allows the user to enable or disable pre-802.3af-standard device detection and powering on the specific port. When pre-std-detect is enabled, power will be delivered on PairA only. Default is disabled.

The `no` form of this command resets the action to default.

Examples

Enabling standard device detection:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet pre-std-detect
```

Disabling standard device detection:

```
switch(config-if)# no power-over-ethernet pre-std-detect
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet priority

```
power-over-ethernet priority {critical | high | low}
no power-over-ethernet priority {critical | high | low}
```

Description

Sets PoE priority for an interface. Specifying critical, high, or low indicates the priority of the interface in the event of power over-subscription. Within the same priority level, higher power-priority line-module ports have higher precedence. With same PoE priority and same line-module priority, lower numbered line-module ports have higher precedence. Per-interface PoE priority is low by default.

The `no` form of this command resets the priority to default PoE priority "low".

Examples

Configuring PoE priority:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet priority critical
switch(config-if)# power-over-ethernet priority high
```

Resetting the PoE priority to default:

```
switch(config-if)# no power-over-ethernet priority high
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet quick-poe

```
power-over-ethernet quick-poe <MODULE-ID>
no power-over-ethernet
```



This command is not available on the 6000 or 6100 Switch Series.



This command is available only on the 4100i Switch Series.

Description

Quick PoE is a feature that provides the ability for the switch to provide power to the connected powered device as soon as switch goes through cold reboot. When quick PoE is enabled on the subsystem PoE port disablement and PD demotion is not allowed. also quick PoE enablement is not allowed if any of the port is disabled on the subsystem. User should not over-subscribe the PoE power when quick PoE is enabled. Quick PoE saved configuration will work irrespective of the configuration change at reboot.

Enables quick PoE feature on the switch or the subsystem level. By default, quick-PoE is disabled for the subsystem.

The `no` form of this command disables quick PoE.

Parameter	Description
<code><MODULE-ID></code>	Specifies module number for quick PoE configuration .

Examples

Enabling and disabling quick PoE:

```
switch(config)# power-over-ethernet quick-poe 1/2
switch(config)# no power-over-ethernet quick-poe 1/2
```

```
switch(config-if)# power-over-ethernet quick-poe 1/1
PoE must be enabled on all interfaces before enabling Quick PoE
```

```
switch(config-if)# power-over-ethernet quick-poe 1/3
All interfaces must use the default assigned class before enabling Quick PoE
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet threshold

```
power-over-ethernet threshold <PERCENTAGE>
no power-over-ethernet threshold <PERCENTAGE>
```

Description

Sets the threshold at which the system will send an excess power consumption notification trap. Default value is 80 percentage.

The `no` form of this command resets the action to default.

Parameter	Description
<code><PERCENTAGE></code>	Excess power consumption trap threshold. Range 1-99.

Examples

Setting the power-over-ethernet threshold:

```
switch(config)# power-over-ethernet threshold 75
```

Resetting the power-over-ethernet threshold to default:

```
switch(config-if)# no power-over-ethernet threshold 75
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

power-over-ethernet trap

```
power-over-ethernet trap  
no power-over-ethernet trap
```

Description

This command enables/disables the SNMP trap generation for PoE related events at system level. PoE trap generation is enabled by default.

The `no` form of this command resets the priority to default PoE priority "low".

Examples

Enabling SNMP trap generation for PoE:

```
switch(config)# power-over-ethernet trap
```

Disabling SNMP trap generation for PoE:


```
switch(config-if)# no power-over-ethernet trap
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

show lldp local

```
show lldp local-device [<INTERFACE-ID>]
```

Description

Displays information advertised by the switch if the LLDP feature is enabled by user.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port

Examples

Showing LLDP local device:

```
switch# show lldp local-device 1/1/10
Local Port Data
=====
Port-ID           : 1/1/10
Port-Desc         : "1/1/10"
Port VLAN ID      : 0

PoE Plus Information

PoE Device Type   : Type 2 PSE
Power Source      : Primary
Power Priority     : low
PSE Allocated Power: 25.0 W
PD Requested Power : 25.0 W
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor

show lldp neighbor [<INTERFACE-ID>]

Description

Displays detailed information about a particular neighbor connected to a particular interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port

Examples

Showing LLDP neighbor information when there is only one neighbor:

```
switch# show lldp neighbor-info 1/1/10

Port : 1/1/10
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description : ArubaOS (MODEL: 325), Version Aruba IAP
Neighbor Chassis-ID : 84:d4:7e:ce:5d:68
Neighbor Management-Address : 169.254.41.250
Chassis Capabilities Available : Bridge, WLAN
Chassis Capabilities Enabled :
Neighbor Port-ID : 84:d4:7e:ce:5d:68
Neighbor Port-Desc : eth0
TTL : 120
Neighbor Port VLAN ID :
Neighbor PoEplus information : DOT3
Neighbor Device Type : TYPE2 PD
Neighbor Power Priority : Unkown
Neighbor Power Source : Primary
Neighbor Power Requested : 25.0 W
Neighbor Power Allocated : 0.0 W
Neighbor Power Supported : No
Neighbor Power Enabled : No
Neighbor Power Class : 5
Neighbor Power Paircontrol : No
Neighbor Power Pairs : SIGNAL
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show power-over-ethernet

```
show power-over-ethernet [member <MEMBER-ID>] [brief]
```

Description

Displays the status information of the full system. Displays the brief status of all port or given port if parameter brief is used. Displays the detailed status of given port.

Parameter	Description
<MEMBER-ID>	Displays the detailed status of given member.
<IFNAME>	Display the detailed status of given port.
brief	Display the brief status of all ports or the given port.

Examples

Showing sample output for show power-over-ethernet :

```
switch# show power-over-ethernet

System Power Status

PoE Power Status           : No redundancy
Operational Power Status   : No redundancy
Total Available Power      : 360 W
Total Configured Power     : 0 W
Total Failover Pwr Avl     : 0 W
Total Redundancy Power     : 0 W
Total Power Drawn          : 0 W
Total Power Reserved       : 0 W
Total Remaining Power      : 360 W
Trap Threshold             : 80 %
Trap Enabled               : Yes
Always-on PoE Enabled      : 1/1
Quick PoE Enabled          : None

Internal Power
      Total Power
PS    (Watts)    Status
-----
1/1   0          Ok
1/2   0          Absent
```

Showing sample output for power-over-ethernet brief per-port:

```
switch# show power-over-ethernet 1/1/1 brief
```

Status and Configuration Information for port 1/1/1

Member 1Power Status
 Available: 370 W Reserved: 55.60 W Remaining: 314.40 W
 Always-on PoE Enabled: 1/1

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2

Showing sample output for power-over-ethernet brief for interface range:

```
switch# show power-over-ethernet 1/1/1-1/1/2 brief
```

Status and Configuration Information for port 1/1/1-1/1/2

Power Status
 Available: 360 W Reserved: 0.00 W Remaining: 360.00 W
 Always-on PoE Enabled: 1/1
 Quick PoE Enabled: None

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
1/1/1	Yes	Low	Off	Usage	0.0 W	0.0 W	Searching	N/A	N/A	N/A
1/1/2	Yes	Low	Off	Usage	0.6 W	0.0 W	Searching	N/A	N/A	N/A

Showing sample output for power-over-ethernet for a missing line card:

```
switch# show power-over-ethernet 1/3 brief
```

Module 1/3 is not physically present.

Showing sample output for power-over-ethernet port when physical interface is not present:

```
switch# show power-over-ethernet 2/1/1
```

Interface 2/1/1 is not present.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

You can manage and monitor the AOS-CX switch through Aruba AirWave. The following benefits and functions include:

- Configuration (partial configuration)
- Device topology
- Immediate and historical trend reports
- Monitoring of the device and user connected to the network.
- Network discovery
- Syslogs and trap receiver

For information about which versions of Aruba AirWave support AOS-CX, see the *AOS-CX Release Notes*.

SNMP support and AirWave

For AirWave to discover and monitor the switch, you must:

- Enable the SNMP services on the switch.
- Configure the SNMP agent to use the SNMP version supported by the management station.

SNMP on the switch

The switch provides SNMP services through the management channel and the data interfaces. Functionality, such as device discovery from NMS, syslog and trap forwarding, can be any channel configured by you.

Although the SNMP server can be enabled on both VRFs (`mgmt` and `default`), only one instance of SNMP can be running. The highest priority is on the `default` VRF.

For example, assume that SNMP is first enabled on the `mgmt` VRF (`snmp-server vrf mgmt`). Then, SNMP is enabled on the `default` VRF (`snmp-server vrf default`) without disabling SNMP on the `mgmt` (using an equivalent `no` form of the command). The `show running-config` command displays both `snmp-server vrf` commands; however, the SNMP instance is running only on the `default` VRF (highest priority).

```
switch# config
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
switch(config)# show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.01.
led locator on
!
!
!
snmp-server vrf default
snmp-server vrf mgmt
```

!
...

Supported features with AirWave and the AOS-CX switch

AirWave supports the following features with the AOS-CX switch:

Device management	Device discovery using SNMPv2C and SNMPv3
	Device dashboards
Monitoring management	Device health attributes (device status/reachability)
	Interface and VLAN management
	Initiates an SSH connection from Aruba AirWave to AOS-CX so that the device outputs from the AOS-CX CLI can be displayed in the Aruba AirWave user interface.
	Firmware versions
	Displays neighbor devices connected to AOS-CX switches
	Device topology
Configuration management	Partial configuration
Alarm management	Alarm triggers (device and interface up/down, new device discoveries, custom event triggers)
	Syslogs and traps
Report management	Device inventory, interface utilization, and device reachability reports
	Summary report of device model, firmware, and boot loader version

Configuring the AOS-CX switch to be monitored by AirWave

Prerequisites

Aruba AirWave is active on the network.

Procedure

1. Enable SNMP on the switch by entering the [snmp-server vrf](#) command.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
```

2. Configure the SNMPv2C community to public by entering the `snmp-server community public` command. In this instance, `public` is a read-only community string.

```
switch(config)# snmp-server community public
```

3. The community-string is used by SNMPv1 and SNMPv2C for unencrypted authentication. SNMPv3 lets you encrypt the authentication mechanism. To enable SNMPv3, enter the `snmpv3 user` and `snmpv3 context` commands.

```
switch(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqBNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=

switch(config)# snmpv3 context Admin
```

For discovering devices in AirWave through the SNMPv3 community, the SNMPv3 context name is not mandatory. Devices can still be discovered in Aruba AirWave without the SNMPv3 context name.

4. Enter the `logging` command for enabling syslog forwarding to a remote syslog server, such as AirWave:

```
switch(config)# logging 10.0.10.2 severity debug
```

5. SNMP traps enable an agent to notify the management station of significant events by way of an unsolicited SNMP message. Enable SNMP traps by entering the `snmp-server host` command:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c vrf default
```

SNMP traps cannot be forwarded from AOS-CX 10.00 switches that have the VRF configured as `mgmt`. Later versions of AOS-CX support SNMP trap forwarding even when the VRF is configured as `default` or `mgmt`.

6. For information on how to add a device for monitoring in the Aruba AirWave user interface, see the documentation for Aruba AirWave.

logging

```
logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}
    [udp [<PORT-NUM>] | tcp [<PORT-NUM>] | tls [<PORT-NUM>]]
    [include-auditable-events] [severity <LEVEL>] [vrf <VRF-NAME>]
logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}
    [tls [<PORT-NUM>]] [auth-mode {certificate | subject-name}]
    [legay-tls-renegotiation] [include-auditable-events] [severity <LEVEL>]
    [vrf <VRF-NAME>]
no logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}
```

Description

Enables syslog forwarding to a remote syslog server.

The `no` form of this command disables syslog forwarding to a remote syslog server.

Parameter	Description
{<IPV4-ADDR> <IPV6-ADDR> <HOSTNAME>}	Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.
[udp [<PORT-NUM>] tcp [<PORT-NUM>]]	Specifies the UDP port or TCP port of the remote syslog server to receive the forwarded syslog messages.
udp [<PORT-NUM>]	Range: 1 to 65535. Default: 514
tcp [<PORT-NUM>]	Range: 1 to 65535. Default: 1470
tls [<PORT-NUM>]	Range: 1 to 65535. Default: 6514
include-auditable-events	Specifies that auditable messages are also logged to the remote syslog server.
severity <LEVEL>	Specifies the severity of the syslog messages: ▪ alert: Forwards syslog messages with the severity of alert (6) and emergency (7). ▪ crit: Forwards syslog messages with the severity of critical (5) and above. ▪ debug: Forwards syslog messages with the severity of debug (0) and above. ▪ emerg: Forwards syslog messages with the severity of emergency (7) only. ▪ err: Forwards syslog messages with the severity of err (4) and above ▪ info: Forwards syslog messages with the severity of info (1) and above. Default. ▪ notice: Forwards syslog messages with the severity of notice (2) and above. ▪ warning: Forwards syslog messages with the severity of warning (3) and above.
auth-mode	Specifies the TLS authentication mode used to validate the certificate. ▪ certificate: Validates the peer using trust anchor certificate based authentication. Default. ▪ subject-name: Validates the peer using trust anchor certificates as well as subject-name based authentication.
legacy-tls-renegotiation	Enables the TLS connection with a remote syslog server supporting legacy renegotiation.
vrf <VRF-NAME>	Specifies the VRF used to connect to the syslog server. Optional. Default: default

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```


Enabling the syslog forwarding of messages with a severity of `err` (4) and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF `lab_vrf`:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)# logging example.com tls auth-mode subject name
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server community

```
snmp-server community <STRING>  
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is a password that controls read access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The `no` form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value `public` is automatically defined.

Parameter	Description
<STRING>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server host

```
snmp-server host <IPv4-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
snmp-server host <IPv4-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
snmp-server host <IPv4-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The **no** form of this command removes the specified trap/inform receiver.



Configuring `snmpv3 informs` is not supported.

Parameter	Description
<IPv4-ADDR>	Specifies the IP address of a trap receiver in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100.
trap version <VERSION>	Specifies the trap notification type for SNMPv1 or v2c. Available options are: v1 or v2c.
inform version v2c	Specifies the inform notification type for SNMPv2c.

Parameter	Description
trap version v3	Specifies the trap notification type for SNMPv3.
user <NAME>	Specifies the SNMPv3 user name to be used in the SNMP trap notifications.
community <STRING>	Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: public.
<UDP-PORT>	Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.
vrf <VRF-NAME>	Specifies the name of the VRF on which to send the notifications.

Examples

```

switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public port 5000
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public port 5000 vrf default

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin port
2000

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server vrf

```
snmp-server vrf <VRF-NAME>
no snmp-server vrf <VRF-NAME>
```

Description

Configures the VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF.

The `no` form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Parameter	Description
<VRF-NAME>	Specifies the VRF on which the SNMP agent listens for incoming requests. The SNMP agent can listen on either the <code>mgmt</code> or <code>default</code> VRF. If configured for both, the SNMP agent listens on <code>default</code> , which has a higher priority.

Example

```
switch(config)# snmp-server vrf default
```

```
switch(config)# no snmp-server vrf default
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 context

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]
no snmpv3 context <NAME> [vrf <VRF-NAME>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The `no` form of this command removes the specified SNMP context.

Parameter	Description
<NAME>	Specifies the name of the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark (?).
vrf <VRF-NAME>	Specifies the VRF associated with the context. Default: default.
community <STRING>	Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user

```
snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass {plaintext | ciphertext}
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass {plaintext | ciphertext} <PRIV-PWORD>] ]
no snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass <PRIV-PWORD>] ]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context.

The **no** form of this command removes the specified SNMPv3 user.

Parameter	Description
<NAME>	Specifies the SNMPv3 username. Range 1 - 32 printable ASCII characters, excluding space and question mark.
auth <AUTH-PROTOCOL>	Specifies the authentication protocol used to validate user logins. Available options are: md5 or sha.
auth-pass {plaintext ciphertext} <AUTH-PWORD>	Specifies the SNMPv3 user password. Range for plaintext is 8 - 32 printable ASCII

Parameter	Description
	characters, excluding space and question mark. Range for <code>ciphertext</code> is 1 - 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.
<code>priv <PRIV-PROTOCOL></code>	Specifies the SNMPv3 security protocol (encryption method). Available options are: <code>aes</code> or <code>des</code> .
<code>priv-pass {plaintext ciphertext} <PRIV-PWORD></code>	Specifies the SNMPv3 user privacy passphrase. Range for <code>plaintext</code> is 8 - 32 printable ASCII characters, excluding space and question mark. Range for <code>ciphertext</code> is 1 - 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

Examples

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
```

Removing an SNMPv3 user named Admin:

```
switch(config)# no snmpv3 user Admin
```

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
```

Copying an SNMP user from switch 1 to switch 2.

On switch 1, configure a user called **Admin**, then issue the `show running-config` command to display switch configuration settings. The `snmpv3 user` command uses the `ciphertext` option to protect the user's passwords.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword
priv des priv-pass plaintext myprivpass
switch1(config)# exit
switch1# show running-config
```

```

Current configuration:
!
!Version ArubaOS-CX TL.10.00.0003-8017-gdeb0606~dirty
!
!
snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNimqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
ssh server vrf mgmt
!
!
!
!
interface mgmt
    no shutdown
    ip dhcp
    vlan 1

```

On switch 2, execute the snmpv3 user command that was displayed by `show running-config` on switch 1. This creates the user on switch 2 with the same configuration settings.

```

switch1(config)# snmpv3 user Admin auth sha auth-pass
ciphertextAQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNimqtfYbJYCgAAALkGFJVcSp3nZ3o=priv
des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing Updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

Documentation Feedback

Aruba is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.